

Fragile Watermarking: Visual Evaluation Report

Generated: May 04, 2026 21:18 | Total Experiments: 32

Experiment 1: Peppers — Attack: Crop 33%



1. Original

2. Watermarked

3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.15 dB	Recovered PSNR	34.49 dB
Watermarked SSIM	0.9861	Recovered SSIM	0.9689
True Positive Rate (TPR)	1.0000	Recovered NC	0.9959
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

Experiment 2: Peppers — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.15 dB	Recovered PSNR	38.05 dB
Watermarked SSIM	0.9861	Recovered SSIM	0.9795
True Positive Rate (TPR)	0.5000	Recovered NC	0.9982
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

Experiment 3: Peppers — Attack: Salt_and_Pepper_1%



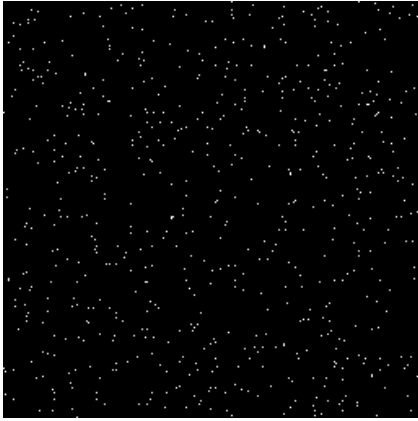
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.15 dB	Recovered PSNR	25.56 dB
Watermarked SSIM	0.9861	Recovered SSIM	0.8417
True Positive Rate (TPR)	0.6197	Recovered NC	0.9679
False Positive Rate (FPR)	0.0482	Tamper F1-Score	0.1949

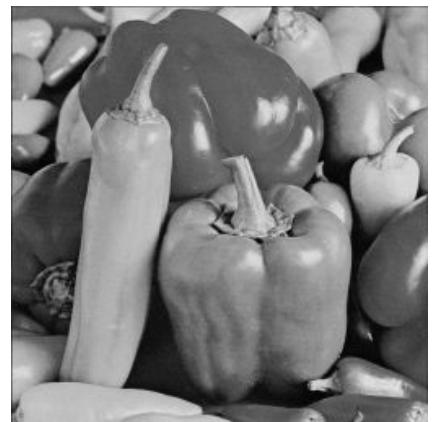
Experiment 4: Peppers — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked

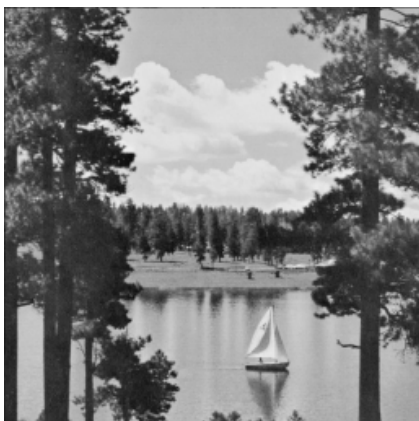


4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.15 dB	Recovered PSNR	37.31 dB
Watermarked SSIM	0.9861	Recovered SSIM	0.9623
True Positive Rate (TPR)	N/A	Recovered NC	0.9979
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A

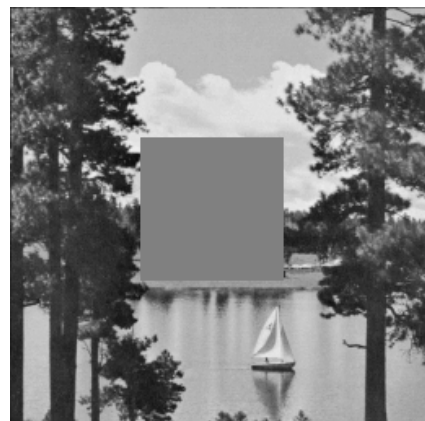
Experiment 5: Sailboat on lake — Attack: Crop 33%



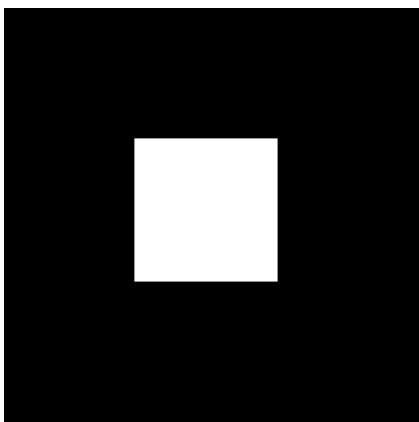
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.15 dB	Recovered PSNR	34.76 dB
Watermarked SSIM	0.9885	Recovered SSIM	0.9710
True Positive Rate (TPR)	1.0000	Recovered NC	0.9974
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

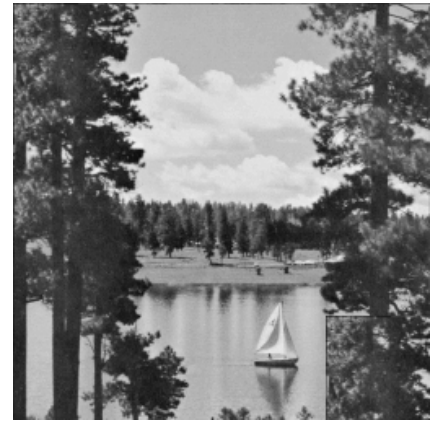
Experiment 6: Sailboat on lake — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



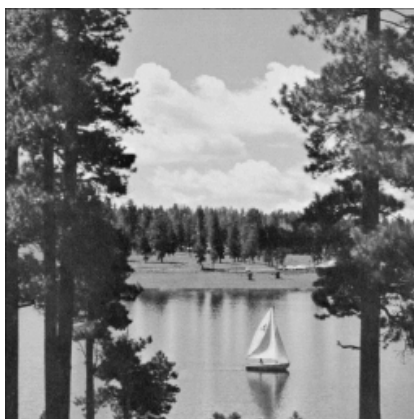
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.15 dB	Recovered PSNR	38.82 dB
Watermarked SSIM	0.9885	Recovered SSIM	0.9810
True Positive Rate (TPR)	0.5000	Recovered NC	0.9990
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

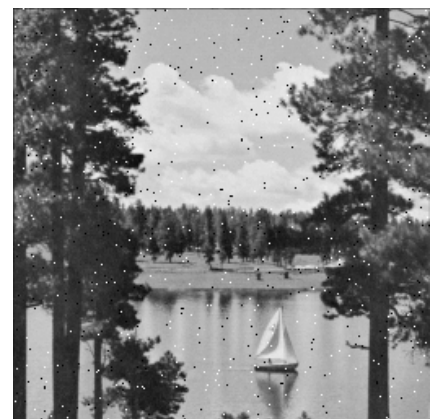
Experiment 7: Sailboat on lake — Attack: Salt_and_Pepper_1%



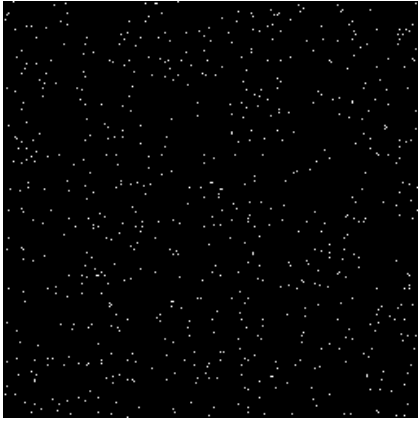
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



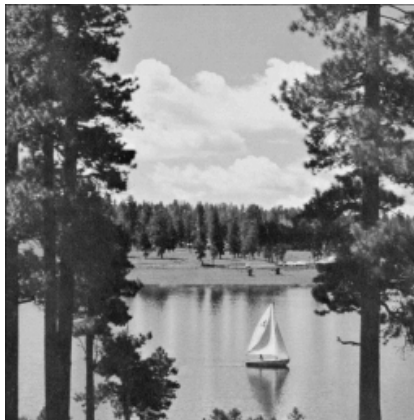
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.15 dB	Recovered PSNR	23.65 dB
Watermarked SSIM	0.9885	Recovered SSIM	0.8304
True Positive Rate (TPR)	0.5902	Recovered NC	0.9663
False Positive Rate (FPR)	0.0475	Tamper F1-Score	0.1921

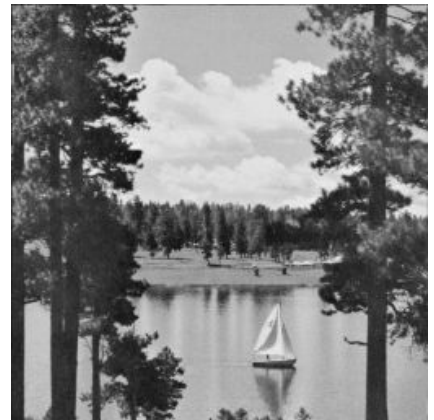
Experiment 8: Sailboat on lake — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

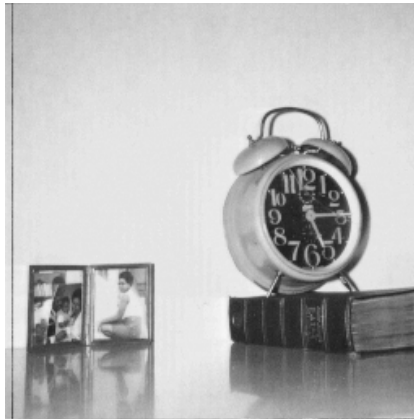
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.15 dB	Recovered PSNR	34.57 dB
Watermarked SSIM	0.9885	Recovered SSIM	0.9563
True Positive Rate (TPR)	N/A	Recovered NC	0.9973
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A

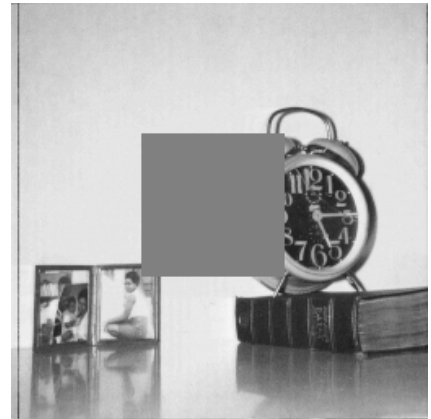
Experiment 9: Clock — Attack: Crop 33%



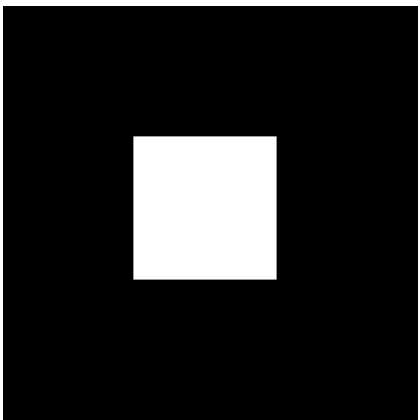
1. Original



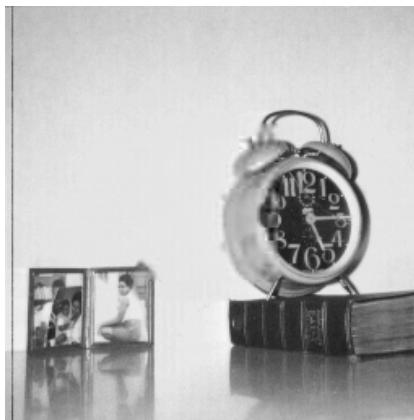
2. Watermarked



3. Attacked



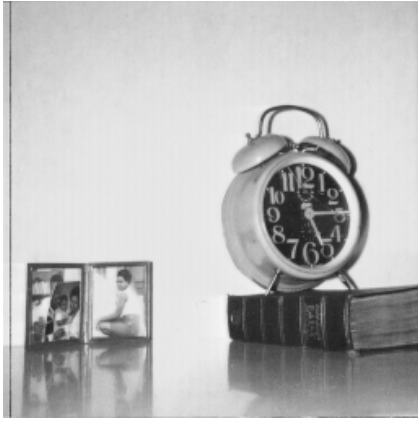
4. True Tamper Map



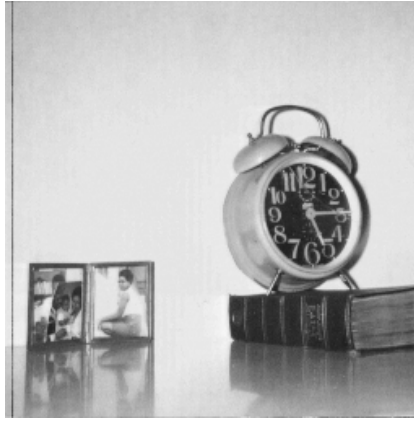
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.23 dB	Recovered PSNR	34.18 dB
Watermarked SSIM	0.9779	Recovered SSIM	0.9682
True Positive Rate (TPR)	1.0000	Recovered NC	0.9962
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

Experiment 10: Clock — Attack: Copy-Move



1. Original



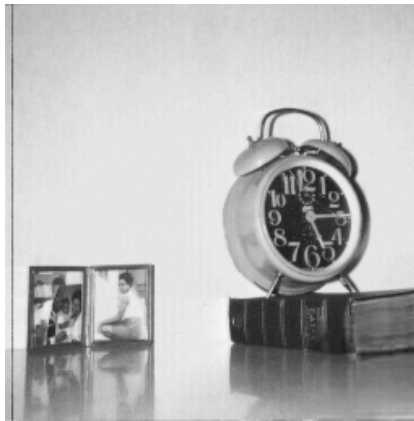
2. Watermarked



3. Attacked



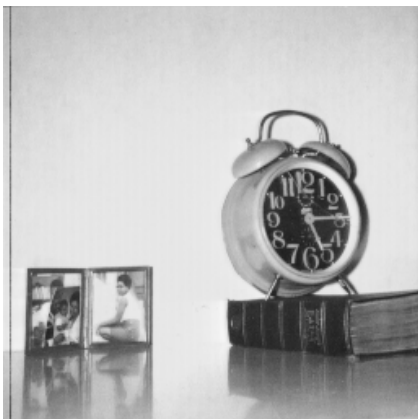
4. True Tamper Map



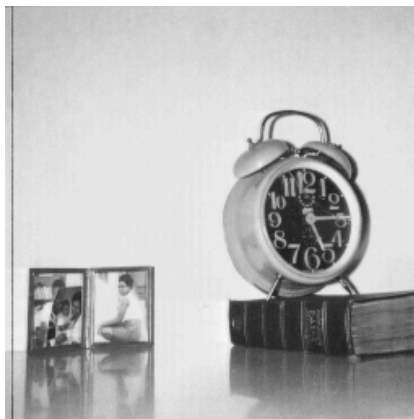
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.23 dB	Recovered PSNR	41.11 dB
Watermarked SSIM	0.9779	Recovered SSIM	0.9711
True Positive Rate (TPR)	0.5000	Recovered NC	0.9992
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

Experiment 11: Clock — Attack: Salt_and_Pepper_1%



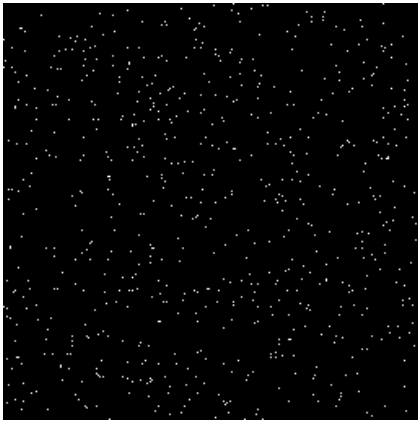
1. Original



2. Watermarked



3. Attacked



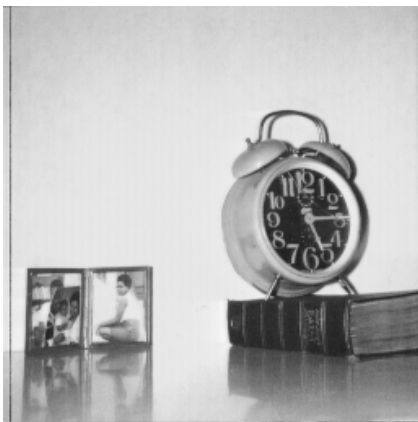
4. True Tamper Map



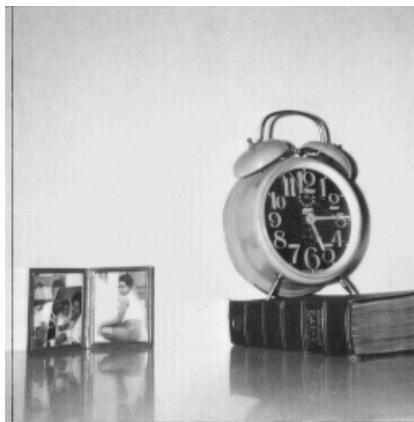
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.23 dB	Recovered PSNR	25.81 dB
Watermarked SSIM	0.9779	Recovered SSIM	0.8539
True Positive Rate (TPR)	0.5757	Recovered NC	0.9739
False Positive Rate (FPR)	0.0470	Tamper F1-Score	0.1942

Experiment 12: Clock — Attack: JPEG_80



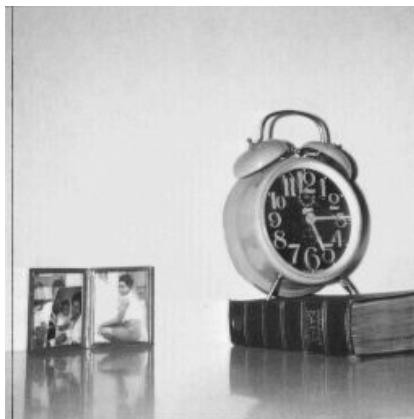
1. Original



2. Watermarked



3. Attacked

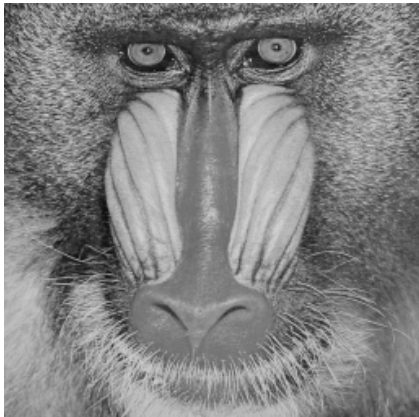


4. True Tamper Map

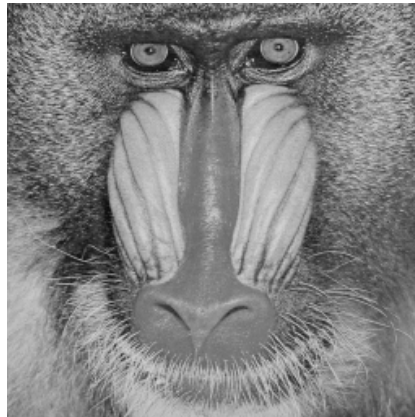
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.23 dB	Recovered PSNR	38.06 dB
Watermarked SSIM	0.9779	Recovered SSIM	0.9665
True Positive Rate (TPR)	N/A	Recovered NC	0.9985
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A

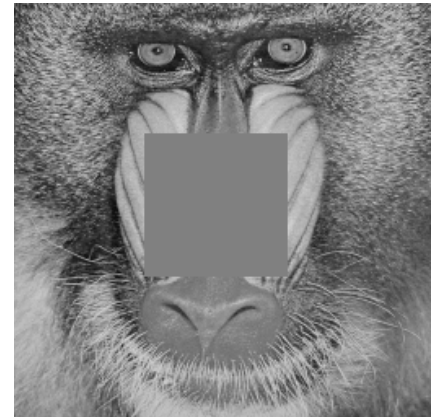
Experiment 13: Mandrill — Attack: Crop 33%



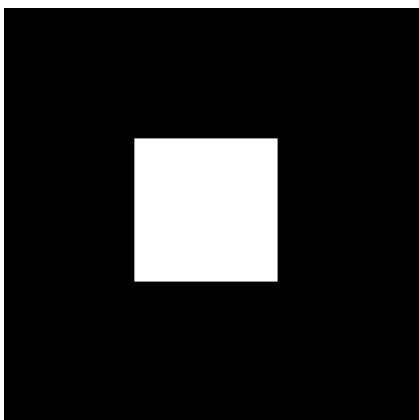
1. Original



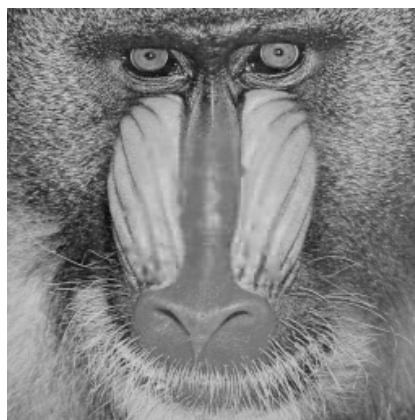
2. Watermarked



3. Attacked



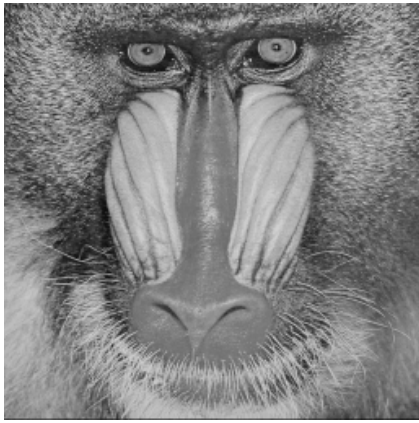
4. True Tamper Map



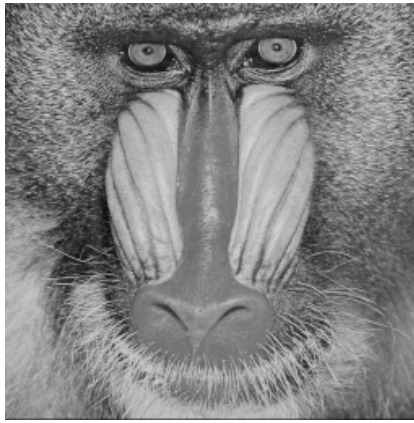
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.13 dB	Recovered PSNR	36.41 dB
Watermarked SSIM	0.9946	Recovered SSIM	0.9702
True Positive Rate (TPR)	1.0000	Recovered NC	0.9950
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

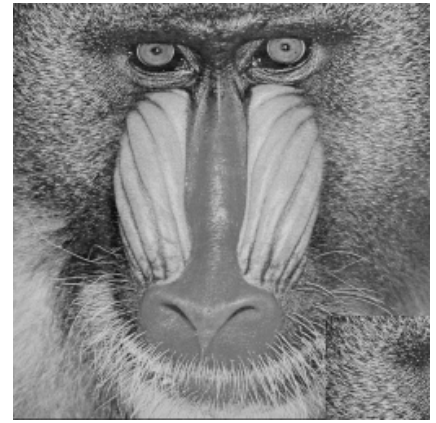
Experiment 14: Mandrill — Attack: Copy-Move



1. Original



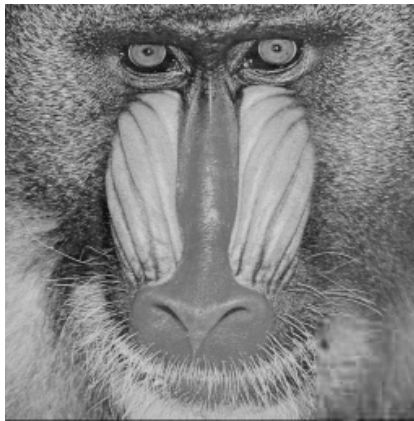
2. Watermarked



3. Attacked



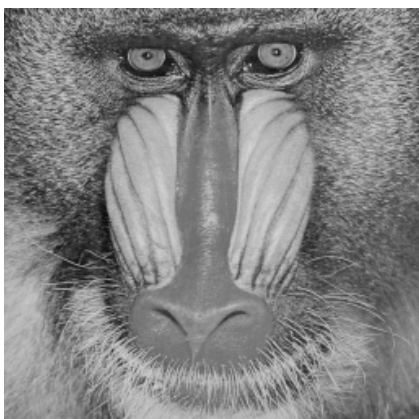
4. True Tamper Map



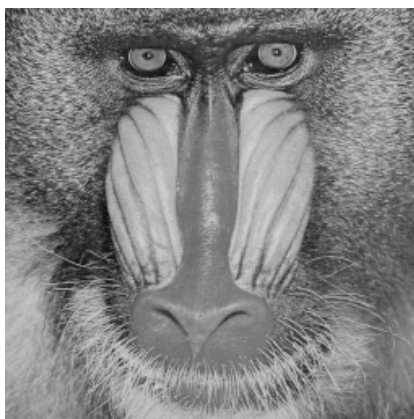
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.13 dB	Recovered PSNR	37.70 dB
Watermarked SSIM	0.9946	Recovered SSIM	0.9782
True Positive Rate (TPR)	0.5000	Recovered NC	0.9963
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

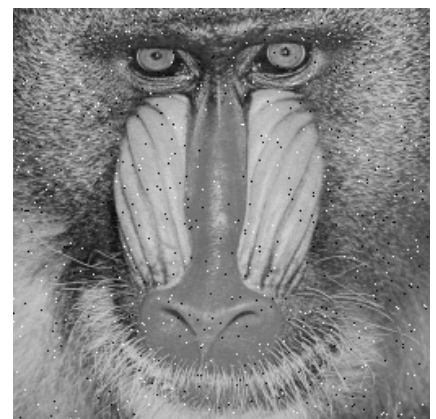
Experiment 15: Mandrill — Attack: Salt_and_Pepper_1%



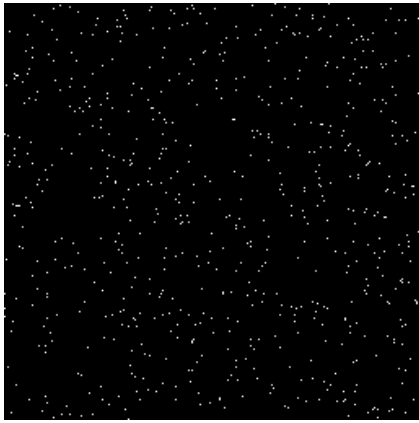
1. Original



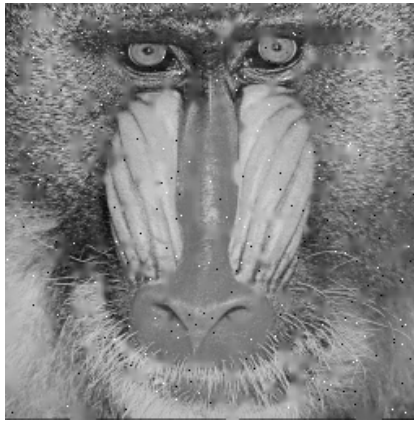
2. Watermarked



3. Attacked



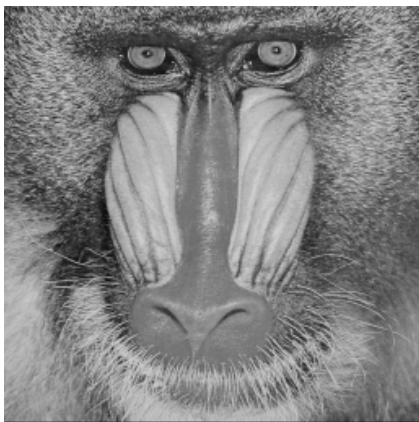
4. True Tamper Map



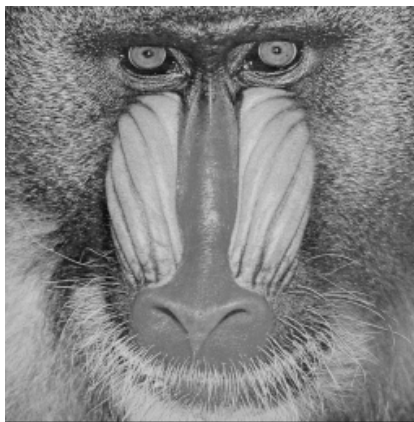
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.13 dB	Recovered PSNR	25.53 dB
Watermarked SSIM	0.9946	Recovered SSIM	0.8290
True Positive Rate (TPR)	0.5842	Recovered NC	0.9384
False Positive Rate (FPR)	0.0460	Tamper F1-Score	0.1936

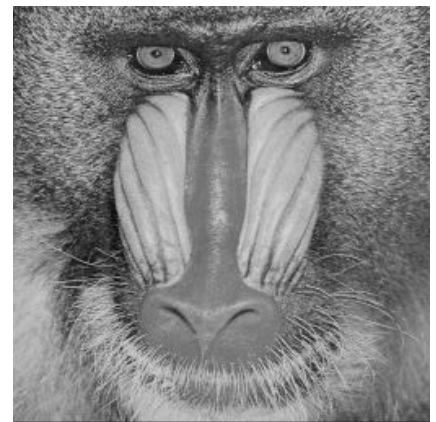
Experiment 16: Mandrill — Attack: JPEG_80



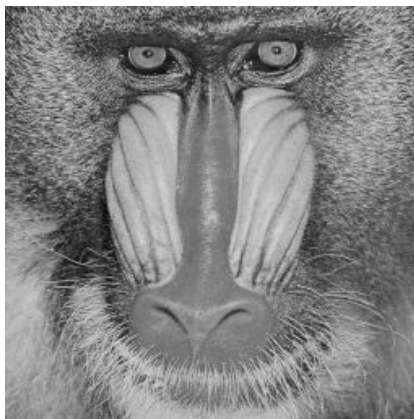
1. Original



2. Watermarked



3. Attacked



5. Recovered Image

4. True Tamper Map

Metric	Value	Metric	Value
Watermarked PSNR	44.13 dB	Recovered PSNR	32.13 dB
Watermarked SSIM	0.9946	Recovered SSIM	0.9438
True Positive Rate (TPR)	N/A	Recovered NC	0.9868
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A

Experiment 17: Walter Cronkite — Attack: Crop 33%



1. Original

2. Watermarked

3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.09 dB	Recovered PSNR	38.90 dB
Watermarked SSIM	0.9774	Recovered SSIM	0.9683
True Positive Rate (TPR)	1.0000	Recovered NC	0.9986
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

Experiment 18: Walter Cronkite — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.09 dB	Recovered PSNR	44.05 dB
Watermarked SSIM	0.9774	Recovered SSIM	0.9786
True Positive Rate (TPR)	0.5000	Recovered NC	0.9996
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

Experiment 19: Walter Cronkite — Attack: Salt_and_Pepper_1%



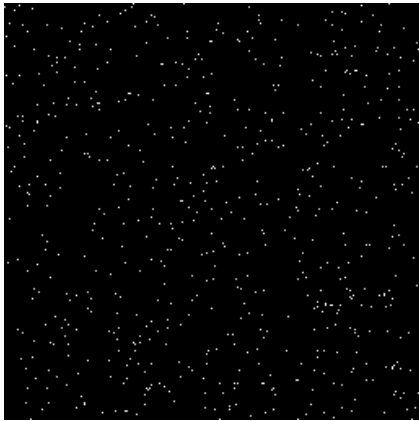
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.09 dB	Recovered PSNR	27.20 dB
Watermarked SSIM	0.9774	Recovered SSIM	0.8596
True Positive Rate (TPR)	0.5991	Recovered NC	0.9790
False Positive Rate (FPR)	0.0436	Tamper F1-Score	0.1981

Experiment 20: Walter Cronkite — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.09 dB	Recovered PSNR	42.32 dB
Watermarked SSIM	0.9774	Recovered SSIM	0.9735
True Positive Rate (TPR)	N/A	Recovered NC	0.9994
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A

Experiment 21: Chemical plant — Attack: Crop 33%



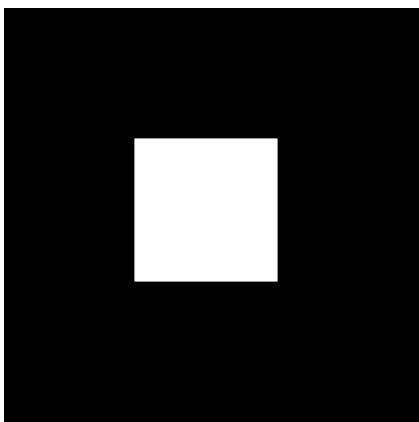
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	33.55 dB
Watermarked SSIM	0.9929	Recovered SSIM	0.9658
True Positive Rate (TPR)	1.0000	Recovered NC	0.9920
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

Experiment 22: Chemical plant — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	36.16 dB
Watermarked SSIM	0.9929	Recovered SSIM	0.9826
True Positive Rate (TPR)	0.5000	Recovered NC	0.9956
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

Experiment 23: Chemical plant — Attack: Salt_and_Pepper_1%



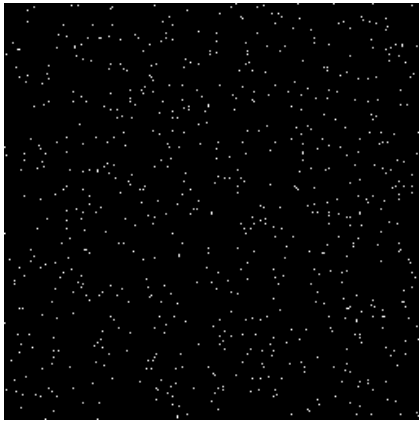
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	24.34 dB
Watermarked SSIM	0.9929	Recovered SSIM	0.7981
True Positive Rate (TPR)	0.6034	Recovered NC	0.9325
False Positive Rate (FPR)	0.0515	Tamper F1-Score	0.1927

Experiment 24: Chemical plant — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked

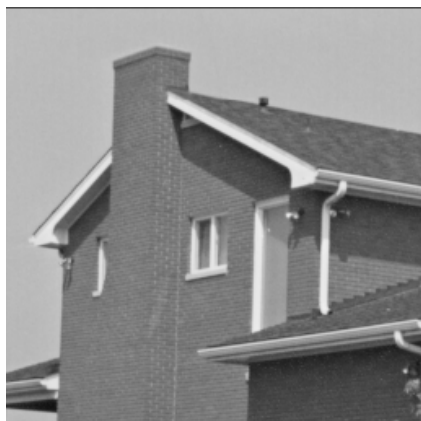


4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	35.15 dB
Watermarked SSIM	0.9929	Recovered SSIM	0.9628
True Positive Rate (TPR)	N/A	Recovered NC	0.9945
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A

Experiment 25: House — Attack: Crop 33%



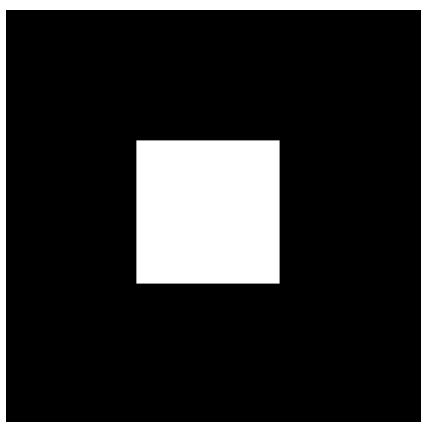
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	35.09 dB
Watermarked SSIM	0.9790	Recovered SSIM	0.9512
True Positive Rate (TPR)	1.0000	Recovered NC	0.9953
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

Experiment 26: House — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



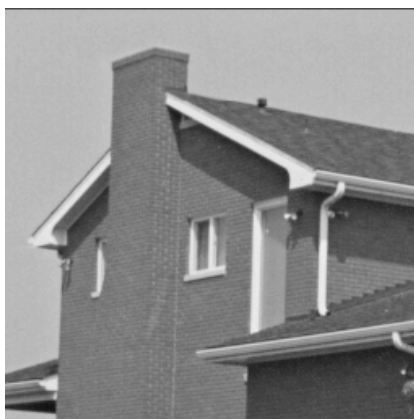
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	39.01 dB
Watermarked SSIM	0.9790	Recovered SSIM	0.9710
True Positive Rate (TPR)	0.5000	Recovered NC	0.9981
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

Experiment 27: House — Attack: Salt_and_Pepper_1%



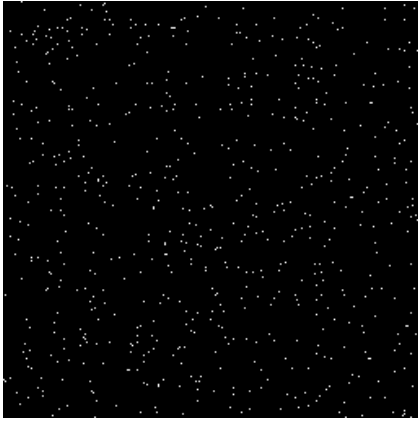
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	26.29 dB
Watermarked SSIM	0.9790	Recovered SSIM	0.8285
True Positive Rate (TPR)	0.5747	Recovered NC	0.9639
False Positive Rate (FPR)	0.0444	Tamper F1-Score	0.1928

Experiment 28: House — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	38.49 dB
Watermarked SSIM	0.9790	Recovered SSIM	0.9470
True Positive Rate (TPR)	N/A	Recovered NC	0.9978
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A

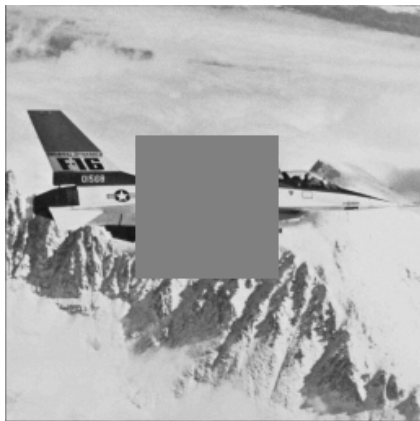
Experiment 29: Airplane — Attack: Crop 33%



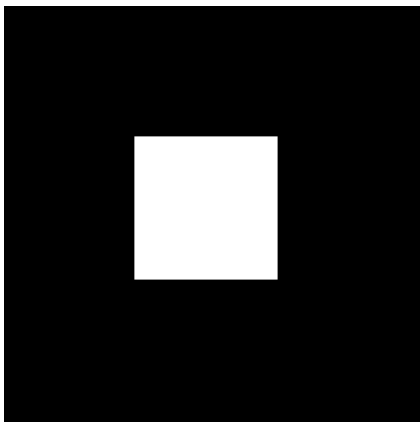
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.17 dB	Recovered PSNR	31.83 dB
Watermarked SSIM	0.9827	Recovered SSIM	0.9592
True Positive Rate (TPR)	1.0000	Recovered NC	0.9896
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

Experiment 30: Airplane — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.17 dB	Recovered PSNR	36.47 dB
Watermarked SSIM	0.9827	Recovered SSIM	0.9731
True Positive Rate (TPR)	0.5000	Recovered NC	0.9965
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

Experiment 31: Airplane — Attack: Salt_and_Pepper_1%



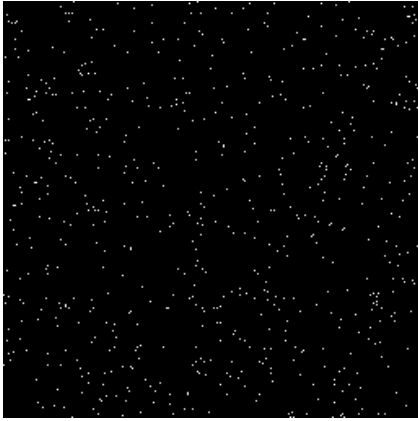
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.17 dB	Recovered PSNR	26.00 dB
Watermarked SSIM	0.9827	Recovered SSIM	0.8623
True Positive Rate (TPR)	0.5552	Recovered NC	0.9609
False Positive Rate (FPR)	0.0425	Tamper F1-Score	0.1920

Experiment 32: Airplane — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.17 dB	Recovered PSNR	36.38 dB
Watermarked SSIM	0.9827	Recovered SSIM	0.9631
True Positive Rate (TPR)	N/A	Recovered NC	0.9964
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A
